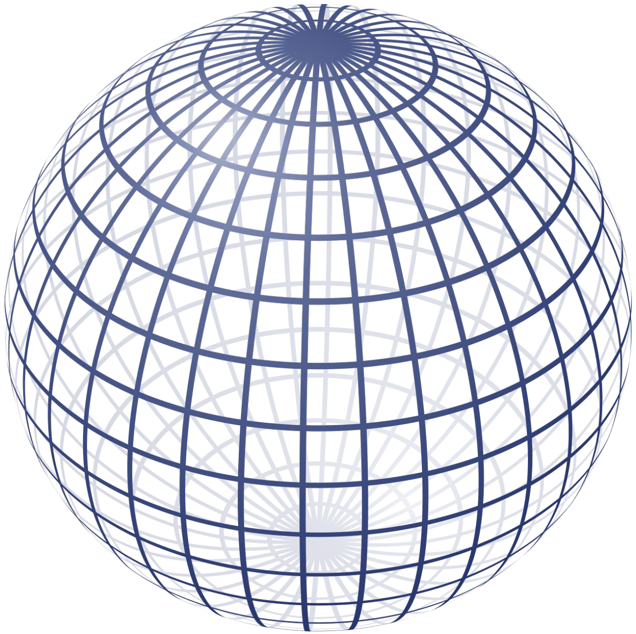
Lecture 5: Calculus & Optimization

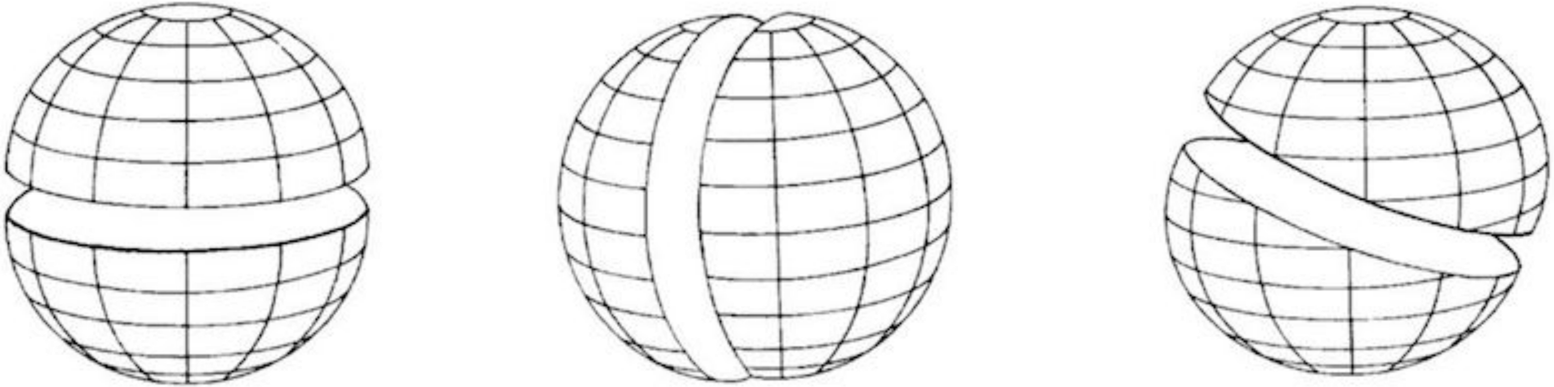
BINF 4002

Borel–Kolmogorov paradox



https://en.wikipedia.org/wiki/Sphere

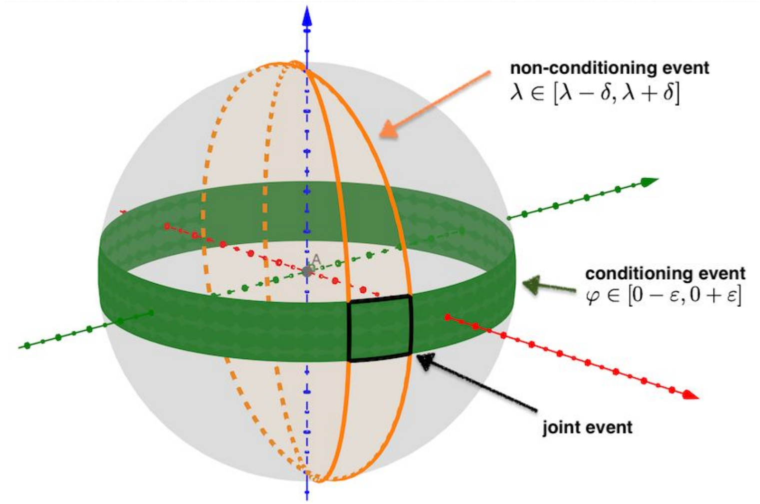
Borel–Kolmogorov paradox

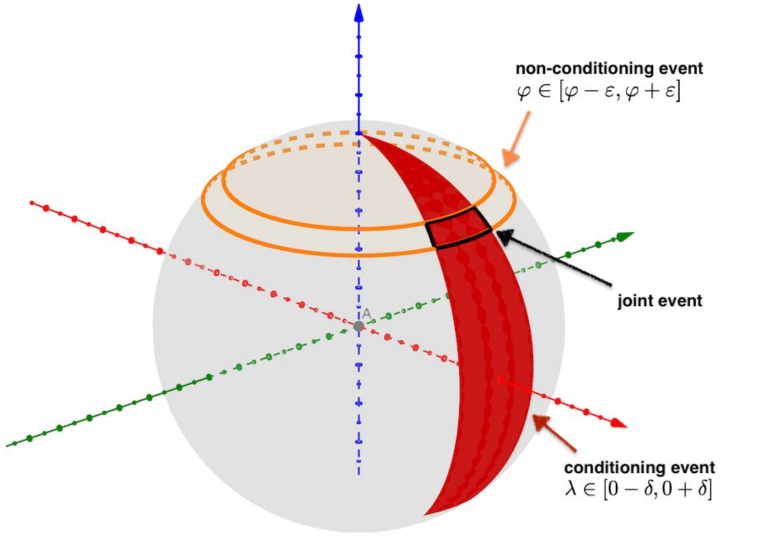


Equator | Meridian/ Longitude line | Another Great Circle

https://gronskiy.com/posts/conditional-probabilities-expectations-and-paradoxes-en/

Borel–Kolmogorov paradox





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Today

1. Logistics Updates (Problem Set 1 & Beyond)
2. Multivariate Functions (and their geometry)
3. The Optimization Problem (and its geometry)

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1. Logistics Updates (Problem Set 1 & Beyond)
2. Multivariate Functions (and their geometry)
    a. Differentiation review!
    b. What is smoothness?
    c. What is the gradient?
    d. What is the tangent surface?
    e. Stationary points
    f. High-dimensionality vs. low-dimensionality
3. The Optimization Problem (and its geometry)
    a. What is an optimization function?
    b. What is an Optima?
    c. Optimization algorithms:
        i. Random search
        ii. Coordinate Descent (and other iterative algorithms)
        iii. Gradient Descent
    d. Optimization landscape challenges: low dimensions
    e. Optimization landscape challenges: high dimensions

Multivariate Functions and Differentiation

Let $f : \mathbb{R}^n \to \mathbb{R}$

Then, we define the partial derivative of f with respect to coordinate x_i (in a basis with the ith basis vector given by $\boldsymbol{e}_i$) evaluated at the point $\boldsymbol{a} \in \mathbb{R}^n$ to be:

$$\left.\frac{\partial f}{\partial x_i}\right|_{\boldsymbol{a}} = \lim_{h \to 0} \frac{f(\boldsymbol{a} + h\boldsymbol{e}_i) - f(\boldsymbol{a})}{h}$$

Derivatives need not exist

- Geometrically, the derivative is how fast the function is changing with respect to the derivative coordinate in an infinitesimally small region about the point.
- If this rate of change is inconsistent depending on the direction you take to navigate the point of interest, the derivative does not exist.

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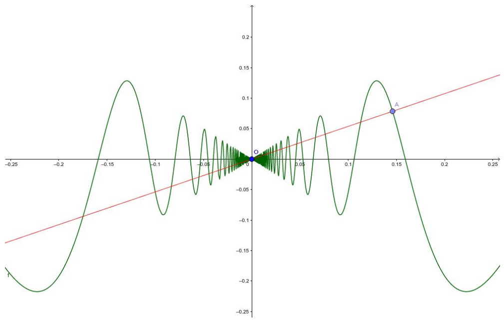
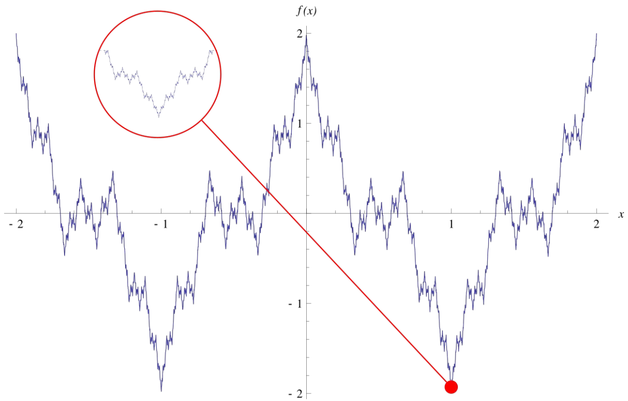
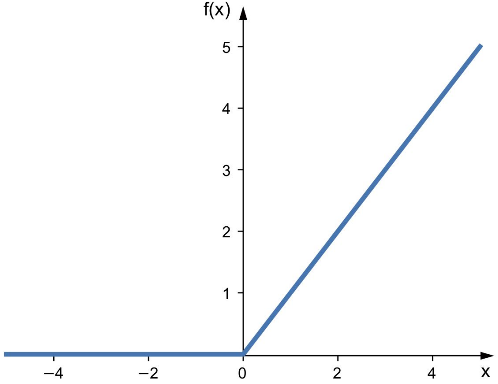


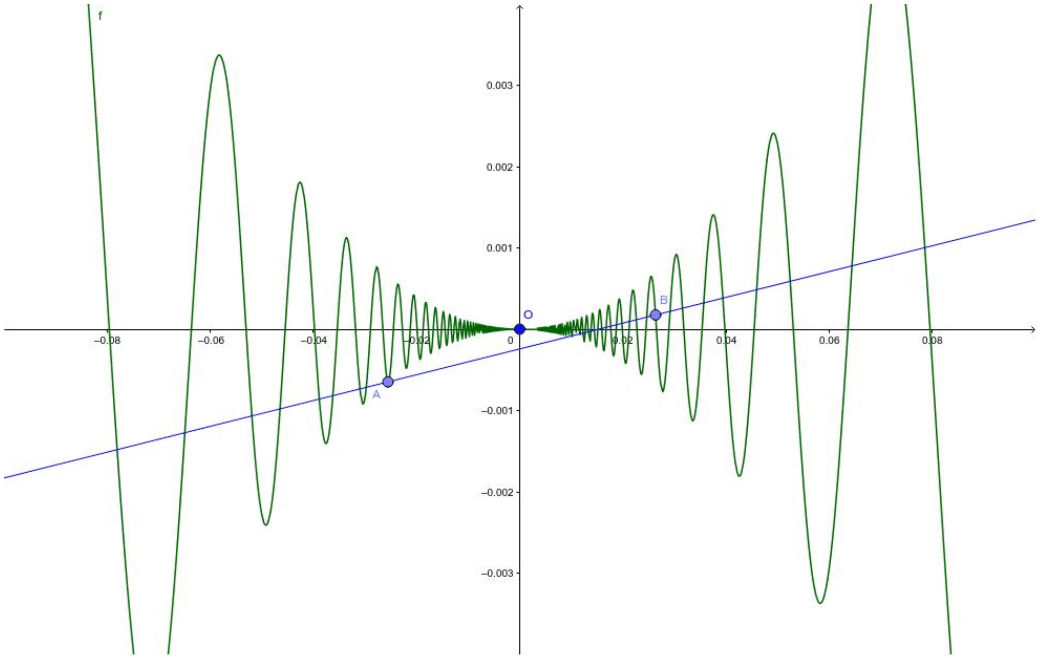




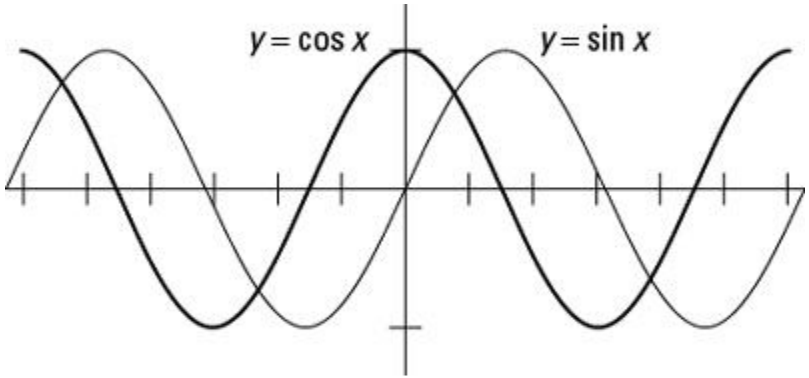
Derivatives need not exist

- Geometrically, the derivative is how fast the function is changing with respect to the derivative coordinate in an infinitesimally small region about the point.
- If this rate of change is inconsistent depending on the direction you take to navigate the point of interest, the derivative does not exist.
- A function is smooth if it is infinitely differentiable (e.g., it is differentiable, and its derivative is differentiable, and its derivative's derivative is differentiable, ...).

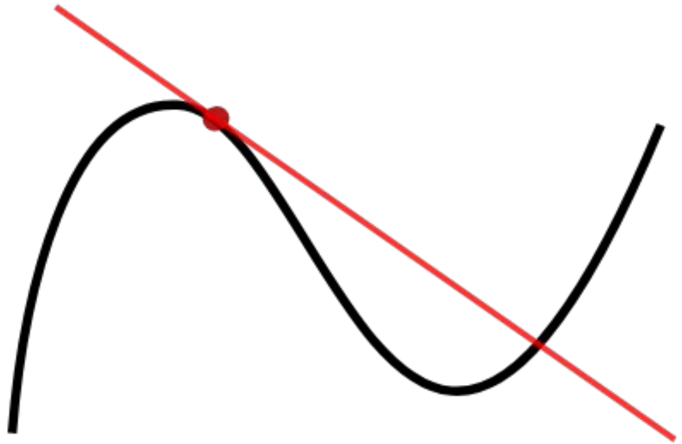
Functions can be differentiable but not smooth



Geometrically, a derivative tells us which direction to perturb the input to increase the output



Geometrically, the tangent surface is the linear function that best approximates the function at the point, and it has slope given by the derivative

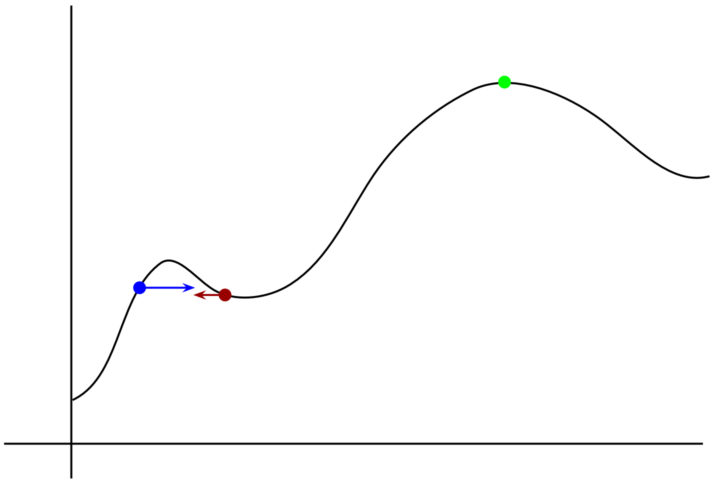


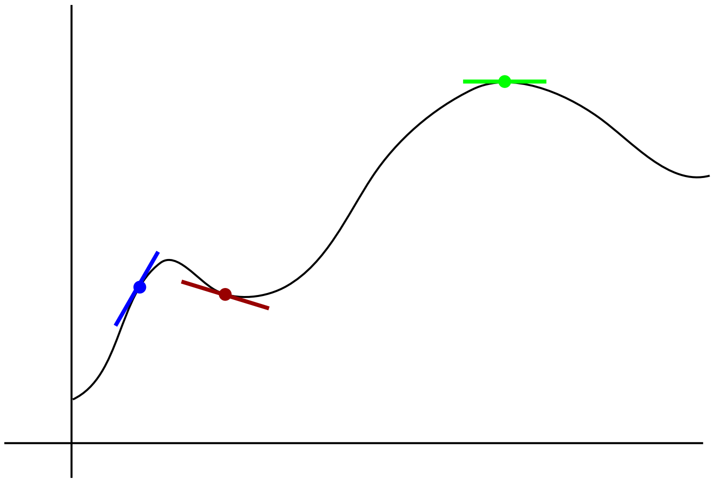
The gradient extends a derivative to a higher-dimensional function, and quantifies the direction of steepest ascent of a real-valued multivariate function.

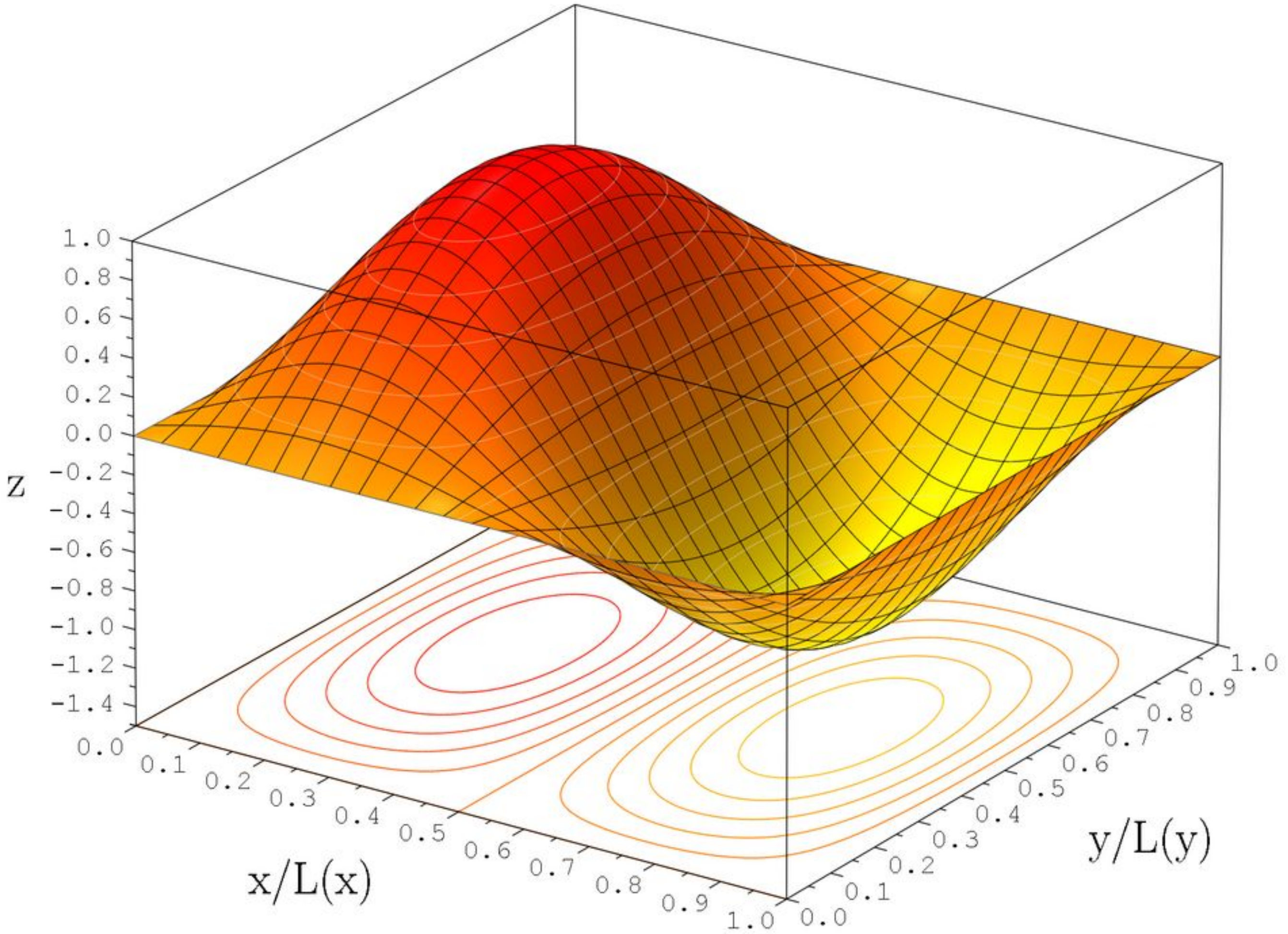
$$\nabla f(p) = \begin{bmatrix} \frac{\partial f}{\partial x_1}(p) \\ \vdots \\ \frac{\partial f}{\partial x_n}(p) \end{bmatrix}$$

The Hessian matrix extends the second derivative to a higher-dimensional function, and quantifies the local curvature of the function.

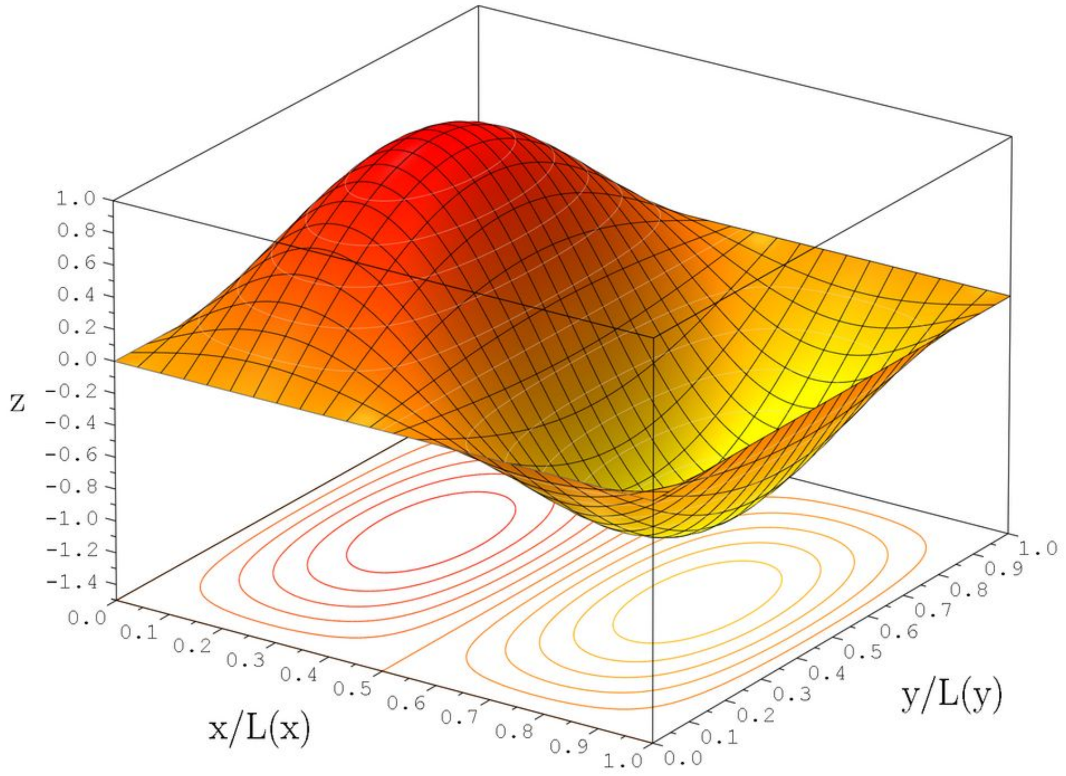
$$\mathbf{H}_f = \begin{bmatrix} \dfrac{\partial^2 f}{\partial x_1^2} & \dfrac{\partial^2 f}{\partial x_1\,\partial x_2} & \cdots & \dfrac{\partial^2 f}{\partial x_1\,\partial x_n} \\[2.2ex] \dfrac{\partial^2 f}{\partial x_2\,\partial x_1} & \dfrac{\partial^2 f}{\partial x_2^2} & \cdots & \dfrac{\partial^2 f}{\partial x_2\,\partial x_n} \\[2.2ex] \vdots & \vdots & \ddots & \vdots \\[2.2ex] \dfrac{\partial^2 f}{\partial x_n\,\partial x_1} & \dfrac{\partial^2 f}{\partial x_n\,\partial x_2} & \cdots & \dfrac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$



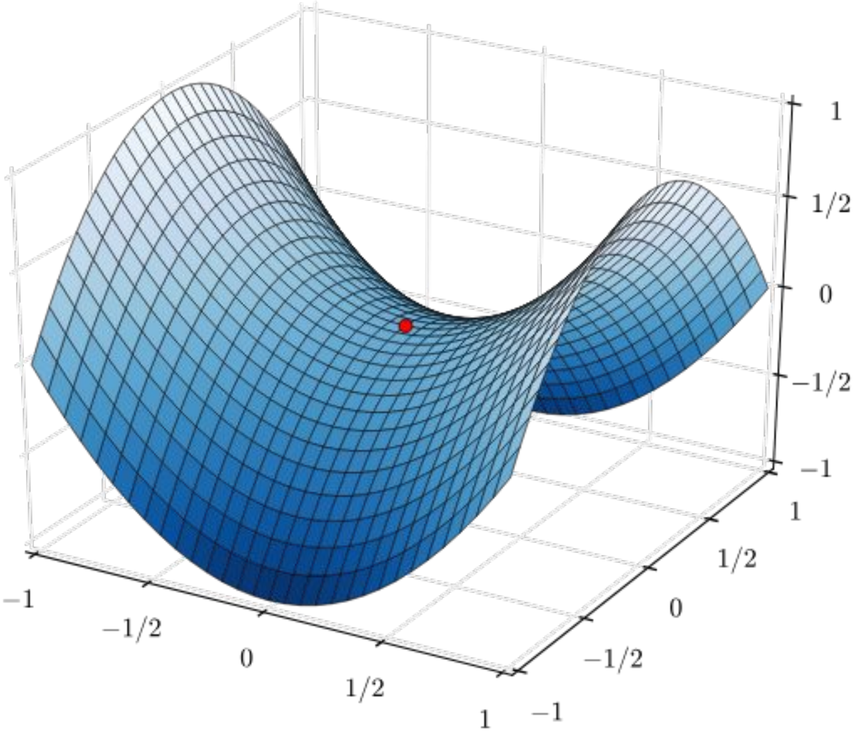



z
1.0
0.8
0.6
0.4
0.2
0.0
-0.2
-0.4
-0.6
-0.8
-1.0
-1.2
-1.4
x/L(x)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
y/L(y)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0

Stationary points have gradient 0 and are locally flat

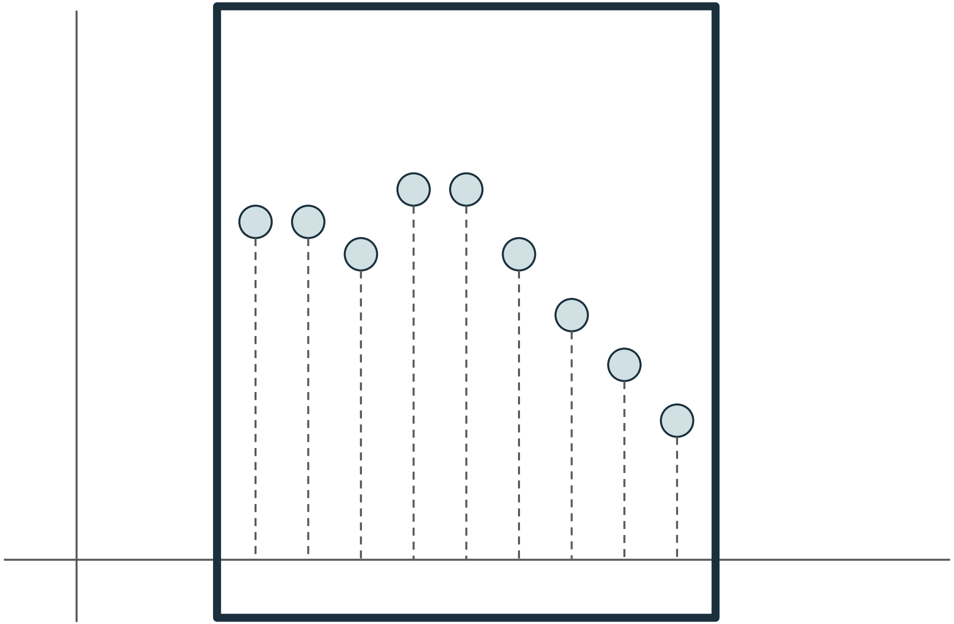


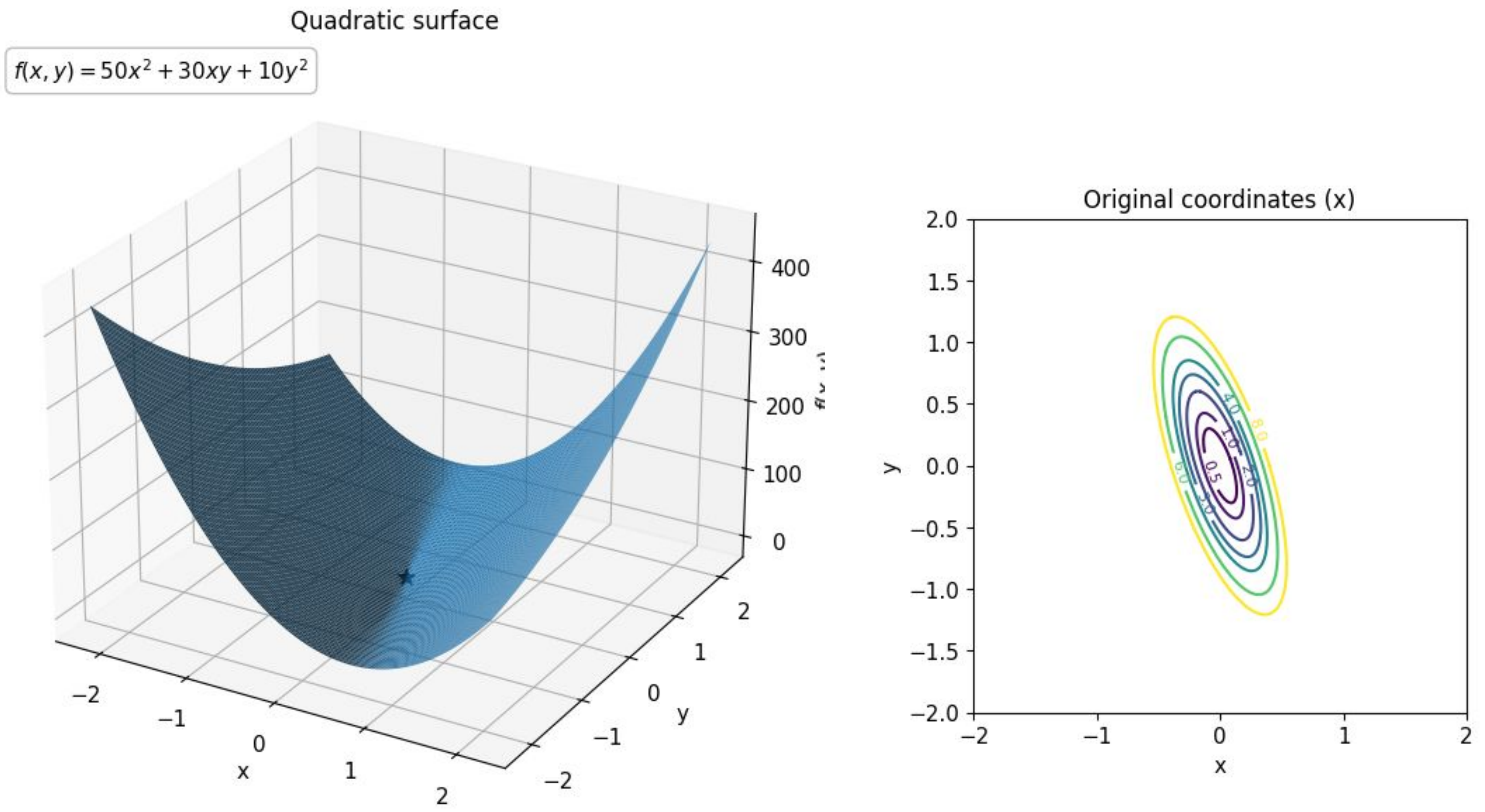
Saddle points are stationary points where the type of critical point differs across coordinates.



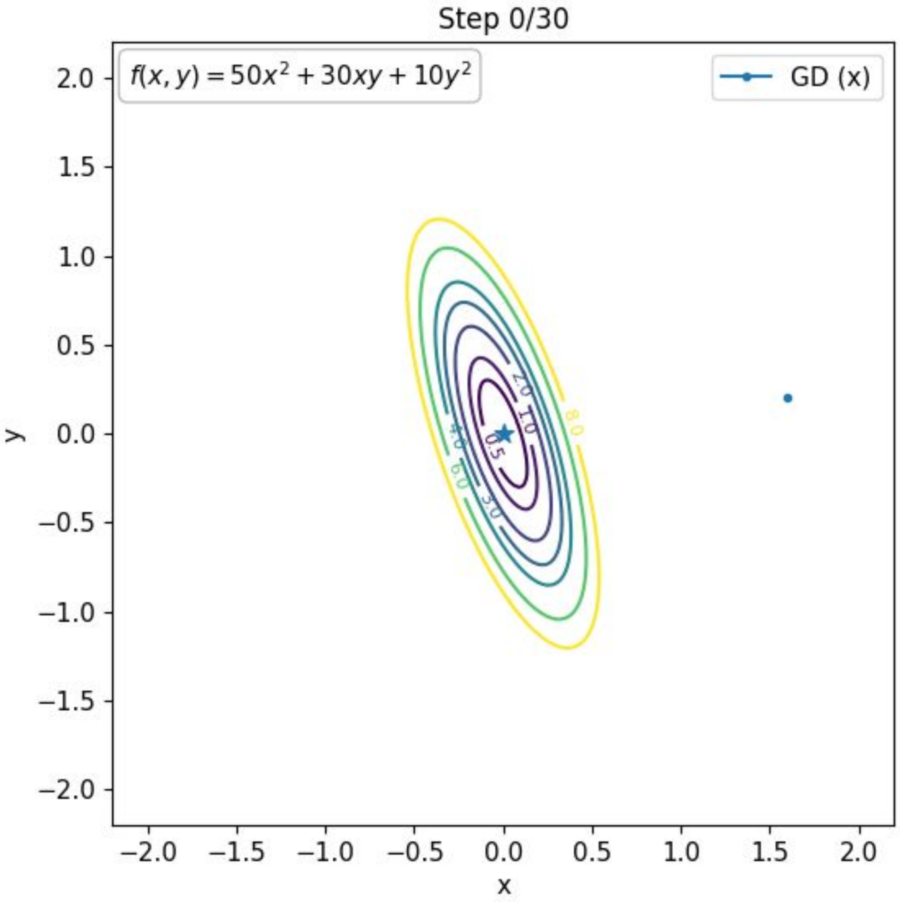
Optimization: How can we identify what parameters will yield a minimal or maximal output value under some function?

Find the optima!





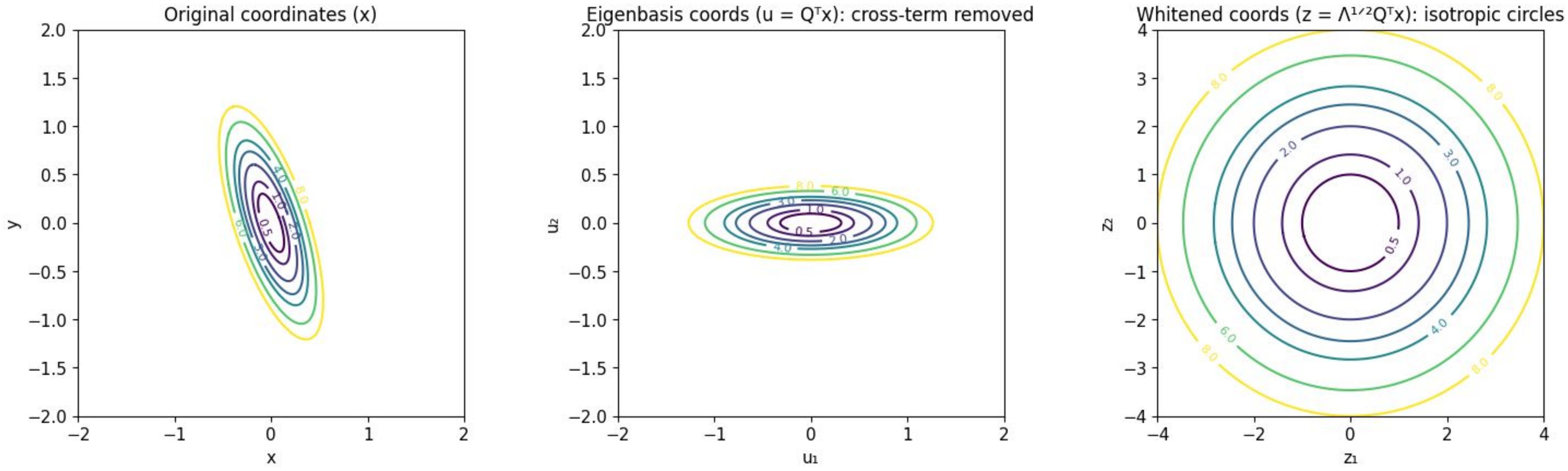
Quadratic surface
$f(x, y) = 50x^2 + 30xy + 10y^2$
x
y
Original coordinates (x)
x
y
0.5
1.0
2.0
4.0
6.0
8.0



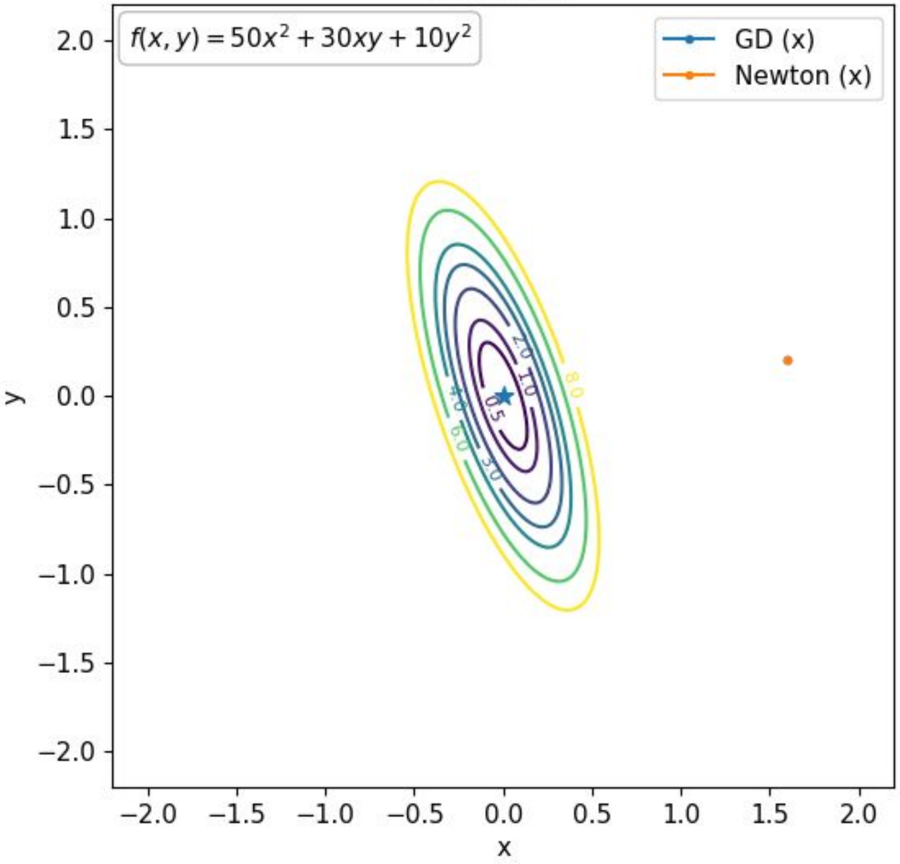
Step 0/30
$f(x, y) = 50x^2 + 30xy + 10y^2$
GD (x)
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
y
x
8.0
6.0
4.0
3.0
2.0
1.0
0.5

The Hessian matrix extends the second derivative to a higher-dimensional function, and quantifies the local curvature of the function.

$$\mathbf{H}_f = \begin{bmatrix} \dfrac{\partial^2 f}{\partial x_1^2} & \dfrac{\partial^2 f}{\partial x_1\,\partial x_2} & \cdots & \dfrac{\partial^2 f}{\partial x_1\,\partial x_n} \\[2.2ex] \dfrac{\partial^2 f}{\partial x_2\,\partial x_1} & \dfrac{\partial^2 f}{\partial x_2^2} & \cdots & \dfrac{\partial^2 f}{\partial x_2\,\partial x_n} \\[2.2ex] \vdots & \vdots & \ddots & \vdots \\[2.2ex] \dfrac{\partial^2 f}{\partial x_n\,\partial x_1} & \dfrac{\partial^2 f}{\partial x_n\,\partial x_2} & \cdots & \dfrac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

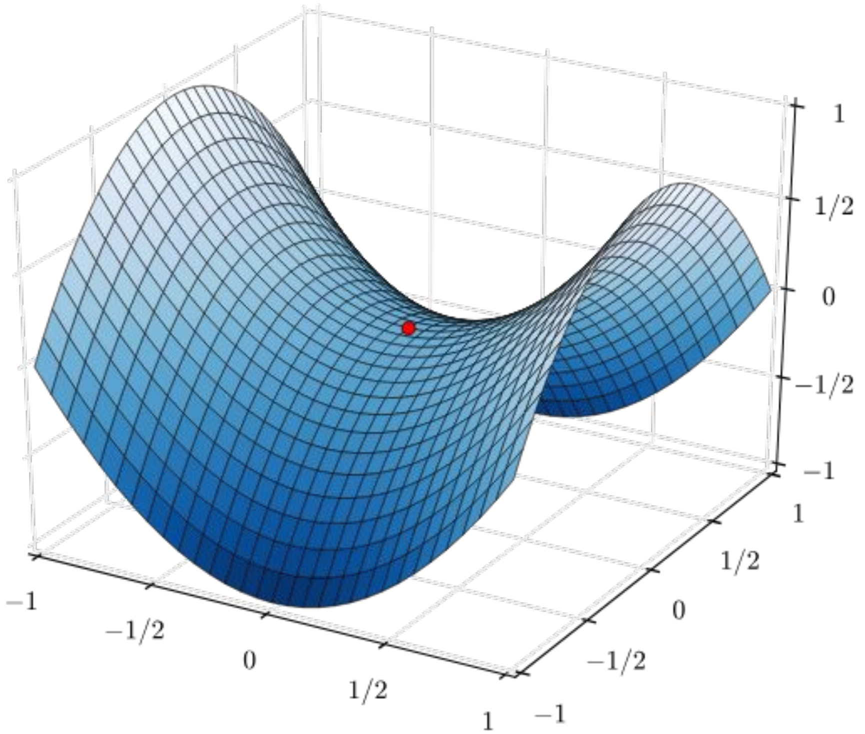

Original coordinates (x)
Eigenbasis coords (u = QᵀX): cross-term removed
Whitened coords (z = Λ^{1/2}Qᵀx): isotropic circles
x
y
u₁
u₂
z₁
z₂
0.5
1.0
2.0
3.0
4.0
6.0
8.0

Step 0/30



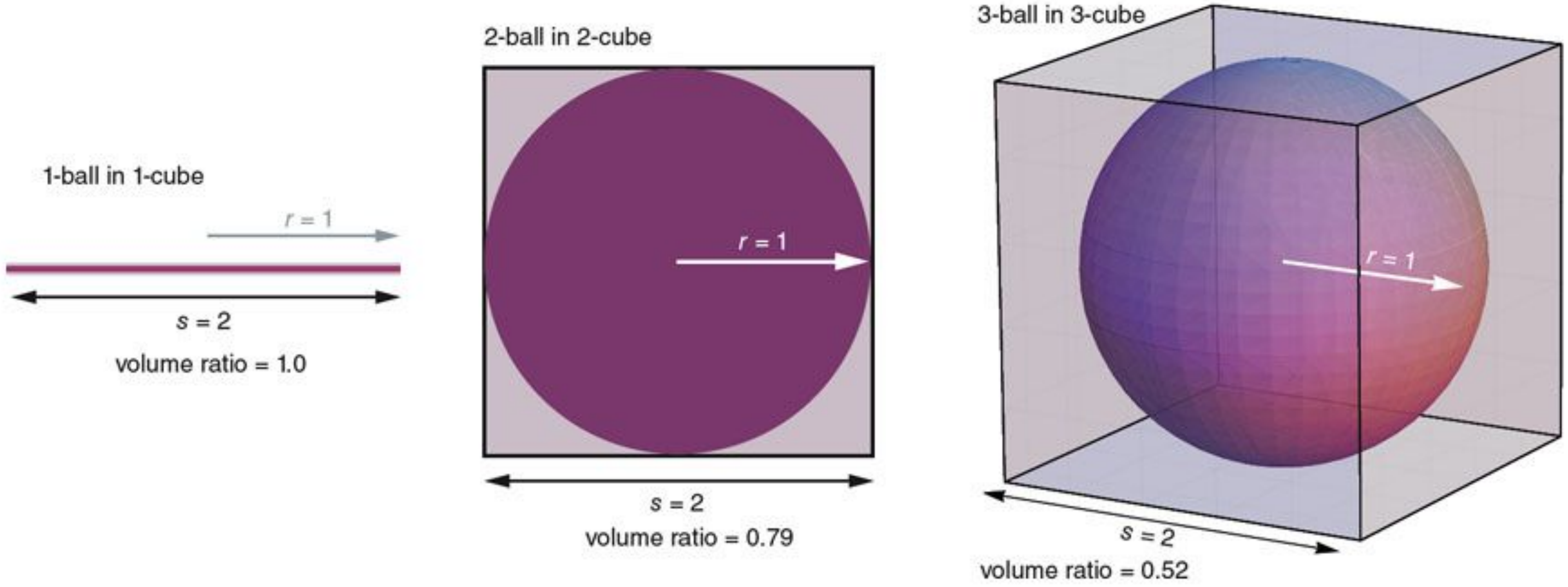
$f(x, y) = 50x^2 + 30xy + 10y^2$
GD (x)
Newton (x)
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
y
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
x
0.5
1.0
2.0
3.0
4.0
6.0
8.0

Saddle points are *unstable* stationary points where the type of critical point differs across coordinates.



High dimensionality

In high dimensions, almost all volume lives outside of neighborhoods around points



Nearly all stationary points are saddle points!

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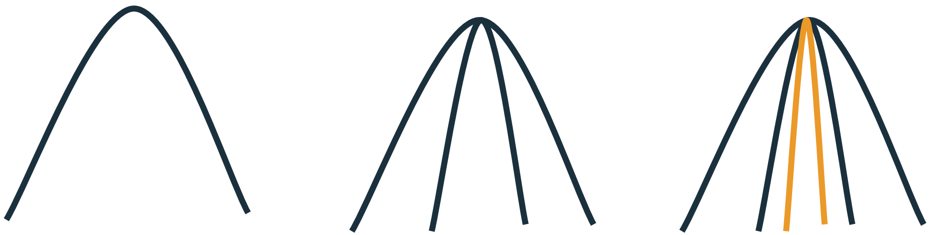

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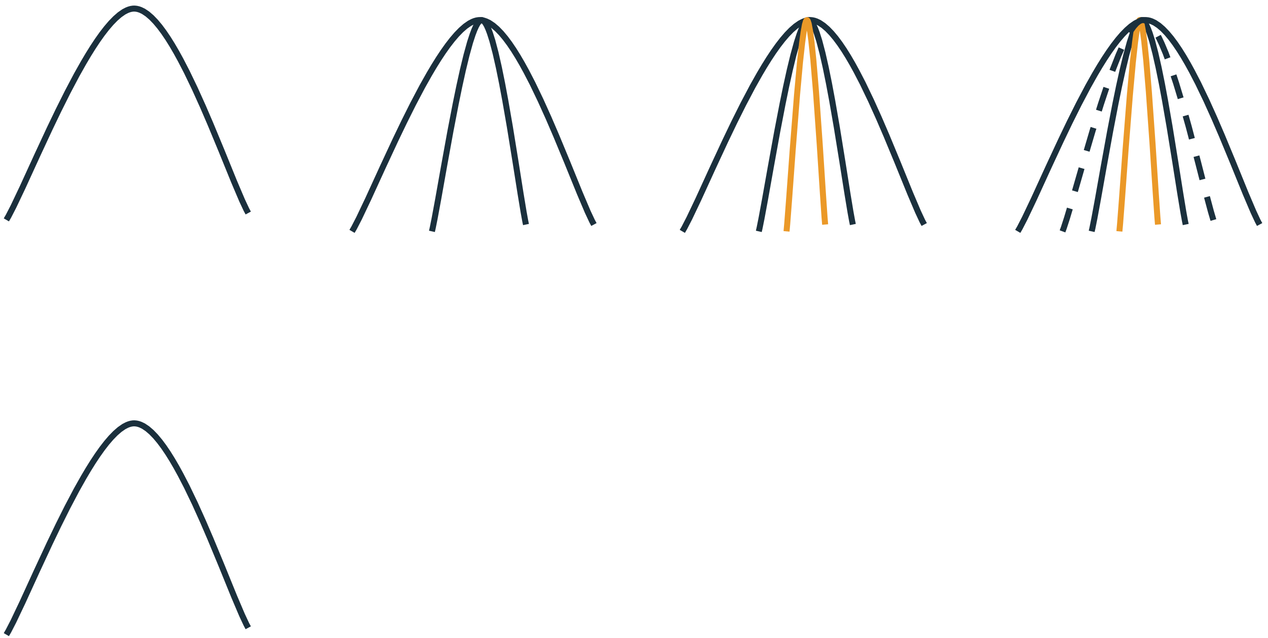

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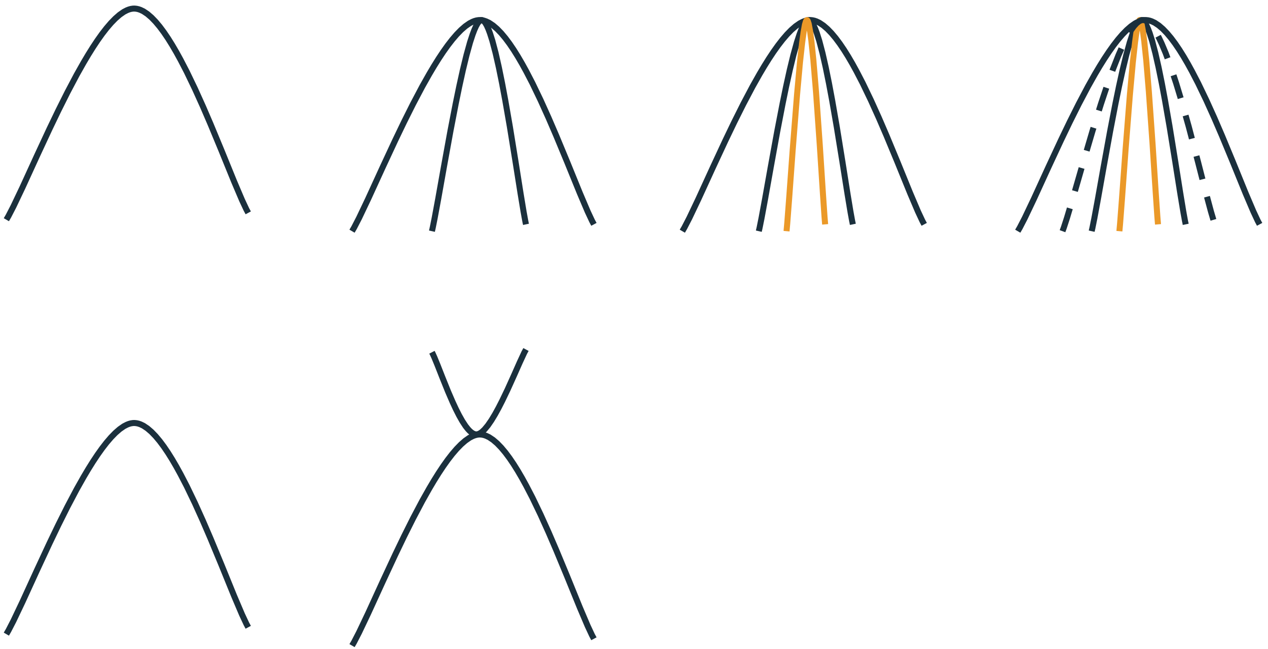

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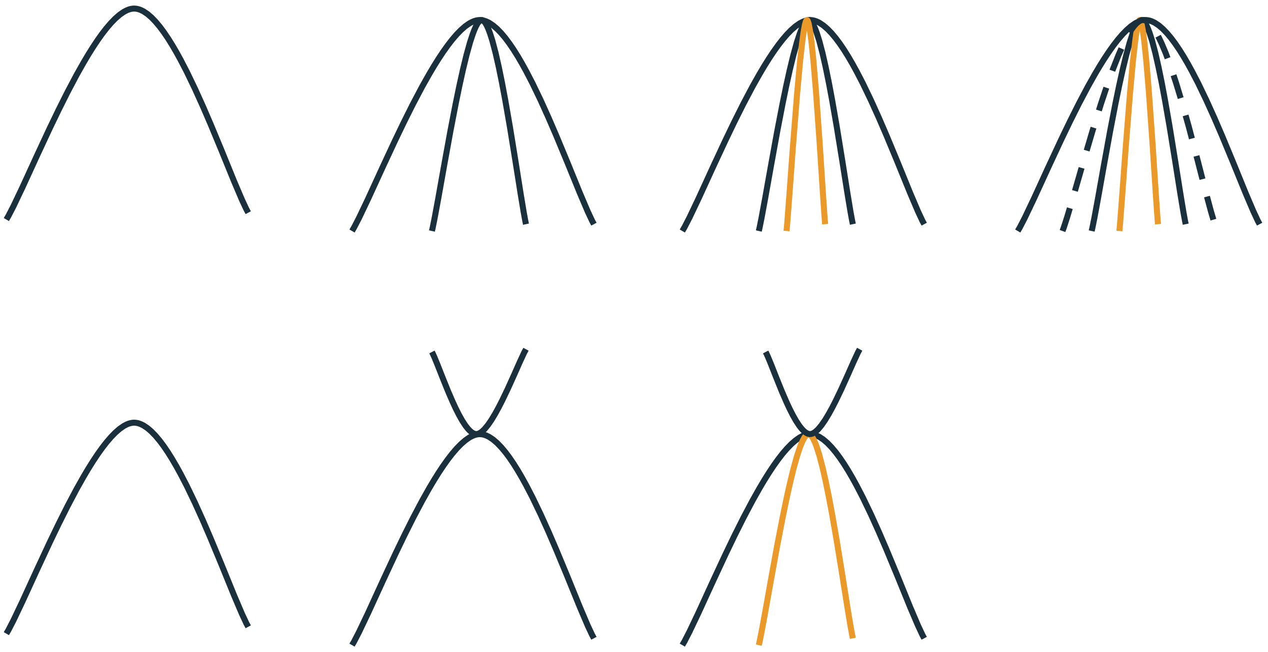

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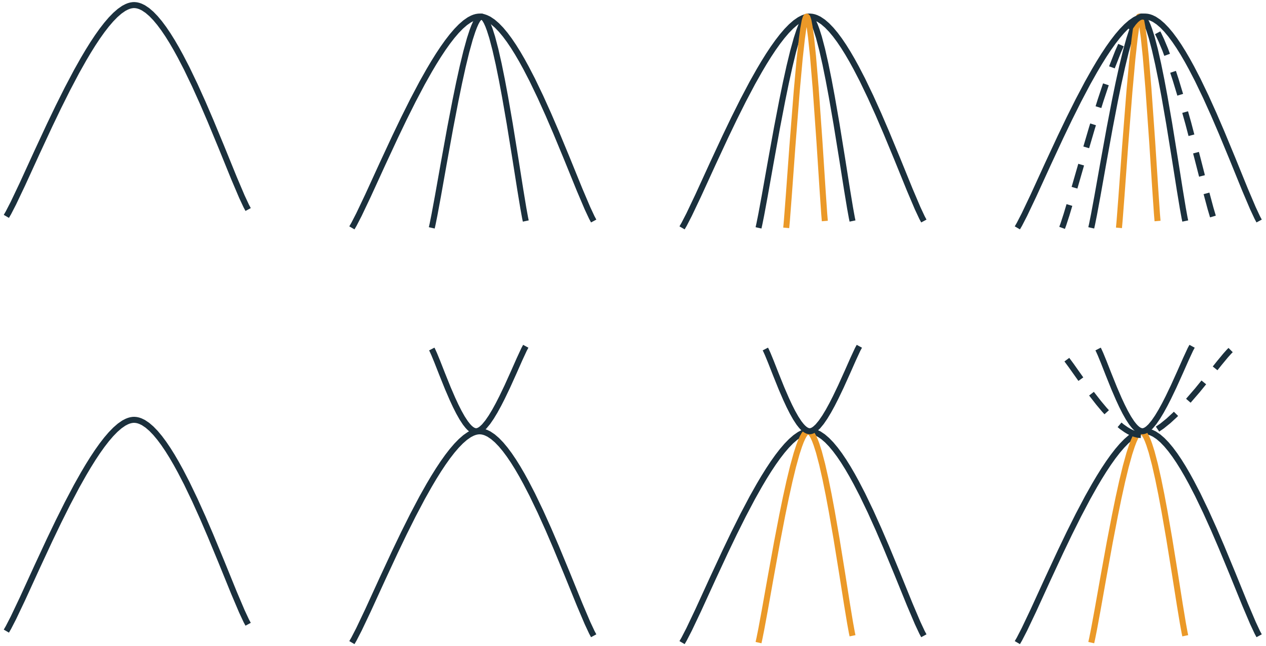

Saddle Points: Unstable Stationary Points

A saddle point is a stationary point where the type of critical point differs across coordinates. It acts as a minimum in one direction and a maximum in an orthogonal direction, creating a characteristic saddle shape and representing an unstable equilibrium.